2.3.1 Friction and the tendency of relative sliding

Friction opposes relative sliding, or the tendency of relative sliding, between contacting surfaces. Sliding friction acts when they slide; static friction acts when the contact does not slide. To find direction, imagine how the surfaces would slide relative to one another without friction. Friction opposes that tendency.

A stationary object on a slope tends to slide down, so friction acts up the slope. A driven tyre pushes the ground backwards and receives forward static friction. For an object rotating with a horizontal platform without slipping, static friction can point towards the centre and provide centripetal force.

2.3.2 Inertia and mass

Inertia is the tendency to maintain rest or constant velocity. Greater mass means greater inertia, so the same resultant force produces a smaller change in velocity per second. Inertia is a property, not an additional force. Without resistance, no continuing forward resultant force is needed to maintain constant velocity.

2.3.3 Newton's first law

In the non-accelerating reference frame used in the textbook, an object with zero resultant force stays at rest or moves at constant velocity in a straight line. The reference frame is the one used by the observer to describe motion. Force changes velocity. Individual forces can balance to give zero resultant force.

2.3.4 Newton's second law

For constant mass, acceleration is determined by the resultant force and points in its direction. A greater resultant force gives greater acceleration for the same mass. A larger mass accelerates less under the same force. Draw a free-body diagram and combine all forces with signs before applying the law.

$$a \propto F_{\text{net}} \quad (m \text{ fixed})$$

$$a \propto \frac{1}{m} \quad (F_{\text{net}} \text{ fixed})$$

$$\sum \vec{F} = m\vec{a}$$

a: acceleration magnitude (m s^{-2}); F_{net}: resultant force magnitude (N); m: mass (kg). Acceleration points along the resultant force.

Resultant force (N), not an individual force; m: constant mass (kg); a: acceleration (m s^{-2}). In an inertial reference frame.

2.3.5 Mass and weight

Mass measures inertia, is expressed in kg and normally remains unchanged when location changes. Weight is gravitational force, measured in N, equal to mass times local gravitational field strength. On the Moon, mass is unchanged but weight is smaller. Weight has direction, and a scale's support-force reading need not equal weight.

$$\vec{W} = m\vec{g}$$

W: weight (N); m: mass (kg); g: gravitational field strength (N kg^{-1}). Weight points along the gravitational field.

2.3.6 Terminal speed

As an object falls through a fluid, increasing speed raises the upward drag. Terminal speed is reached when drag plus upthrust balances weight, making resultant force and acceleration zero while the object continues falling. Only when upthrust is negligible can the balance be simplified to drag equals weight.

2.3.7 Newton's third law

Two interacting objects exert simultaneous forces of equal magnitude and opposite direction on each other. The forces act on different objects, so they do not cancel in one object's free-body diagram. They cancel as internal forces only when both objects are included in a single system.